

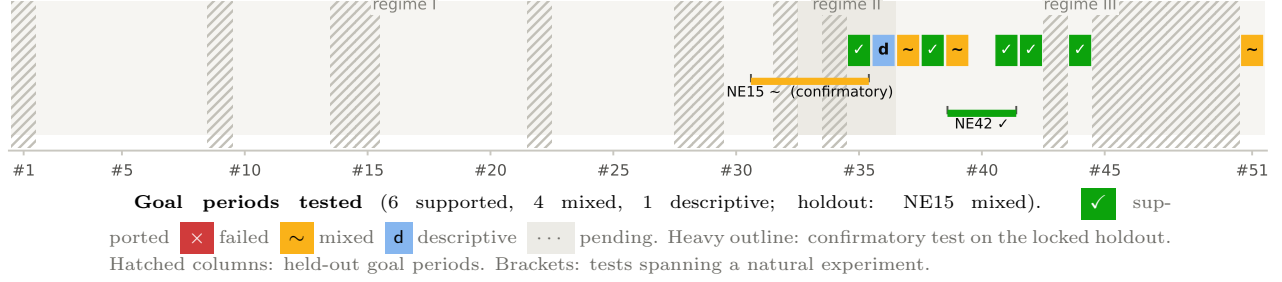
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Rooms cut chat coupling only through what agents read; work is not room-structured, and the holdout verdict, on the buggy table, stays inconclusive.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

A kinetic-Ising fit asks whether rooms make agents coupled blocks, on 99 non-holdout days (#35–#42, #44, #51 to 09-04). Round 1b uses the fixed activity table and day-edge-trimmed nulls. Within-room talk coupling beats cross-room coupling in 6/6 regime-III two-room windows (Fisher $p = 6 \times 10^{-7}$). The per-period test passes in G38, G41, G42 and G44 (mixed G37, G39). The pooled two-way-FE effect of co-location on talk has $z = 4.1$. Ledger read counts absorb it: with reads in the model, co-location falls to $z = 0.2$. Activity spins show no room structure. Entropy production stays at the cross-day null in 15/18 windows, so it is unmeasurable here. **The holdout NE15** (03-16 split) passes the primary pair DiD: talk $\kappa_x -0.022 [-0.038, -0.008]$, $p_{\text{perm}} = 0.02$, and J_{out} falls to ≈ 0 . The pooled TWFE misses ($z = 1.4$), so the verdict is **inconclusive**. Unpredicted, J_{in} rises about $\times 6$. The post-split window (#35) is not held out.

PHYSICAL MODEL

Model 02 (kinetic Ising). The spin $s_i(t) = \pm 1$ says whether agent i talks (talk spin) or acts (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. The field h_i is the daily schedule; a cross-day surrogate removes it. r_i is the room. J_{in} and J_{out} are the same- and cross-room couplings; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g is the Aguilera-Ito-Kolchinsky lower bound on the entropy production σ , with dual weights θ . $\hat{R}$ and D are block correlations and variances. α_{ij} and γ_d are pair and day effects.

$$P(s'_i | s) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

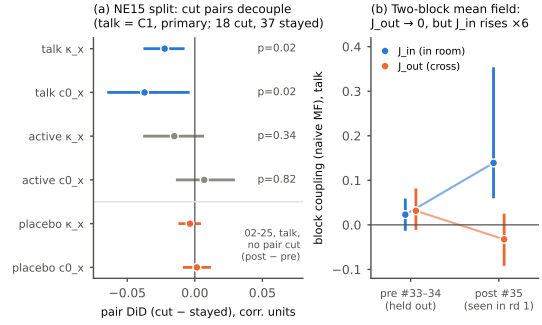
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) The confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 separated pairs minus that of the 37 pairs that stay together (pre #33–34, held out; post #35, not held out). Talk coupling drops (C1 passes); activity does not. The 02-25 placebo separates no one and shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , and within-room J_{in} rises (unpredicted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A room is a read filter. Agents that do not read each other's chat stop timing their chat together, which is close to a restatement of the scaffold. To decouple two agents, stop them reading each other: with read counts known, co-location adds nothing ($z = 0.2$). A pair DiD on 1-min talk spins detects a cut with 18 cut pairs and 5 post-split days, at only 25–60% power. Activity, the actual work, is *not* room-structured. Rooms reorganize conversation, not labor. The $\times 6$ rise in within-room coupling suggests that attention is conserved and moves to roommates (HH90, now in H18). Entropy production is unmeasurable here and has no practical use.

